Lec 17: Performance modeling of computer networks

Graduate Systems (CSE638)

Instructor: Rinku Shah

These slides are inspired and adapted from:

1. *Computer Networking: A Top-Down Approach*
   6th edition, Jim Kurose, Keith Ross, Pearson, 2017



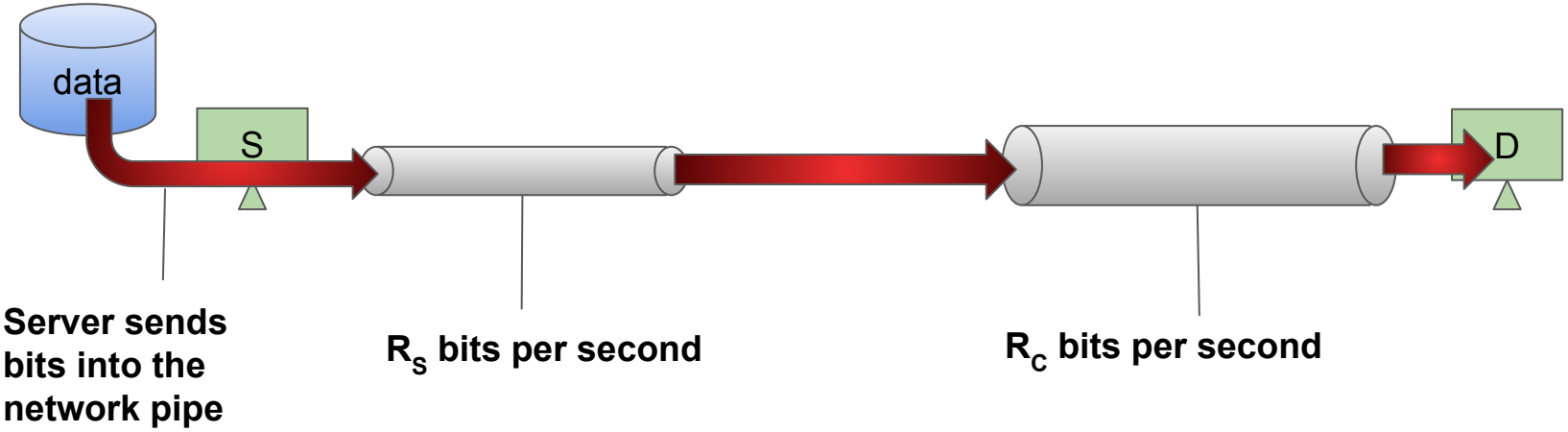
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Modeling the performance of computer networks

Two main attributes to measure network performance

- **Throughput**
  - Rate at which bits are transferred between the sender and receiver
- **Delay**
  - Time taken for a bit to reach from the sender to receiver

Throughput



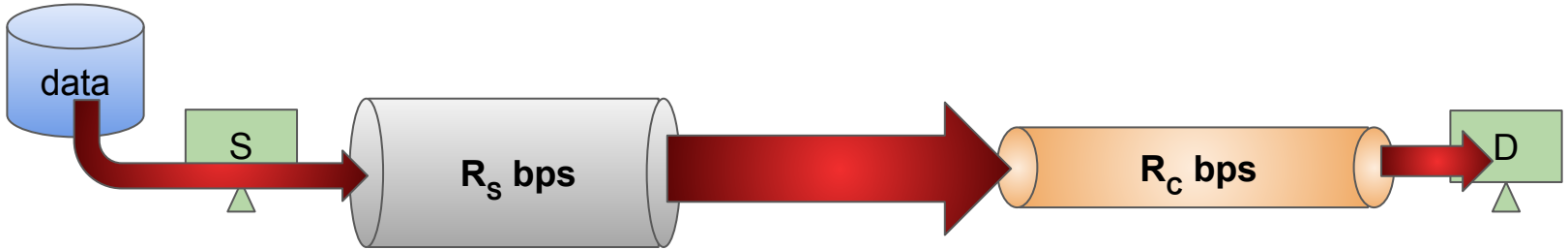
Throughput: Rate at which bits are transferred between the sender and receiver (bits per second)

Throughput (contd.)

$R_s < R_c$ What is the average end-end throughput?

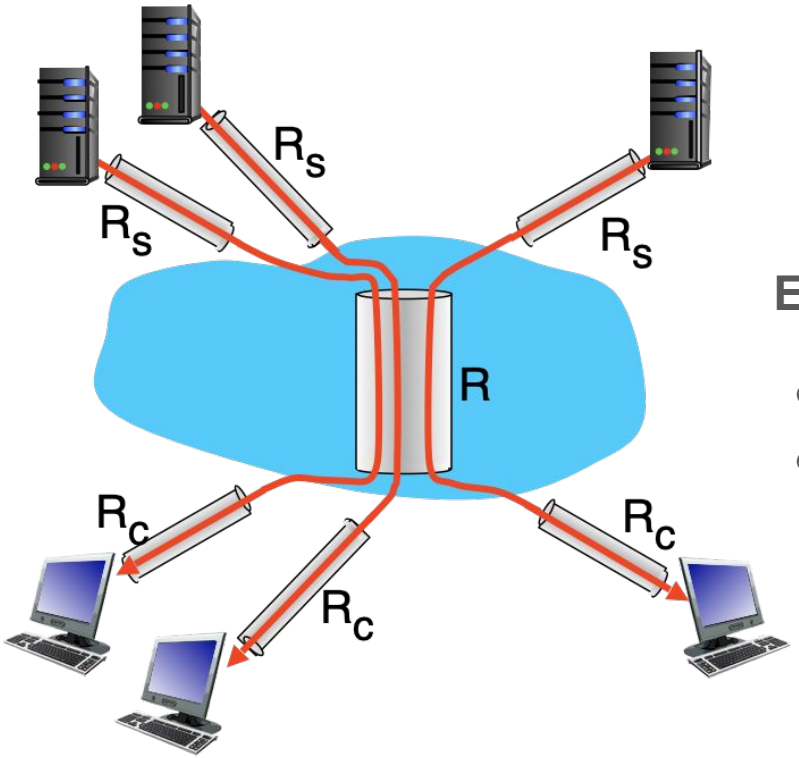


$R_s > R_c$ What is the average end-end throughput?



Bottleneck link on the path constraints end-end throughput

Throughput: Internet scenario



10 connections (fairly) share backbone bottleneck link R bits/sec

End-to-end throughput per connection?

- *min(R_S, R_C, R/10)*
- Typical bottleneck, R_C or R_S

Average throughput NOT ALWAYS = Bottleneck bandwidth

Reasons

- Protocol may add delays
  - Sender has to wait for an ACK from receiver (flow control)
- Packets losses may lead to retransmissions
  - You may not always send useful data
  - GOODPUT: Number of useful bits per sec

Instantaneous throughput may be the bottleneck rate;

Average throughput may be lower!

Throughput calculation

Throughput = Amount of data sent / Time taken

Example:

A file of size `L' bits takes `d' seconds to be transferred.

Average throughput = L/d bps

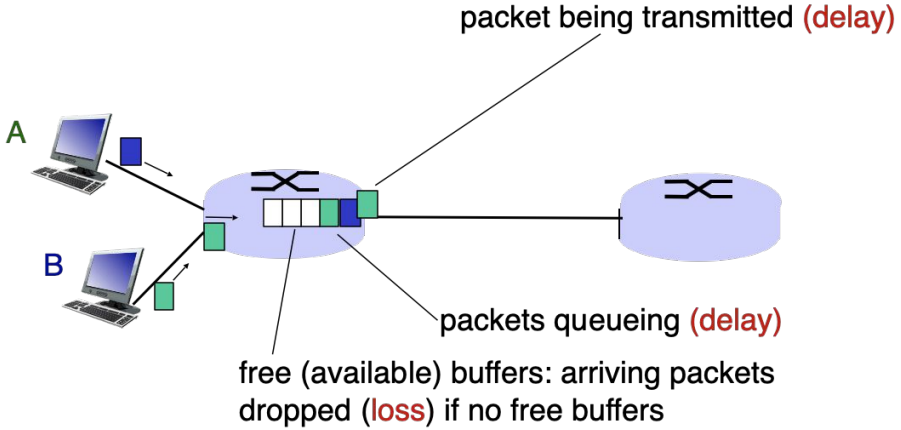
Understand difference between bottleneck bandwidth, instantaneous, and average throughput

Problem

1. Consider a 125KB file that needs to be sent through a network path. The bottleneck bandwidth is 1 Mbps. Suppose the sender continuously sends data at the bottleneck rate, and no packets are lost, no retransmissions. How long does it take to send the file?

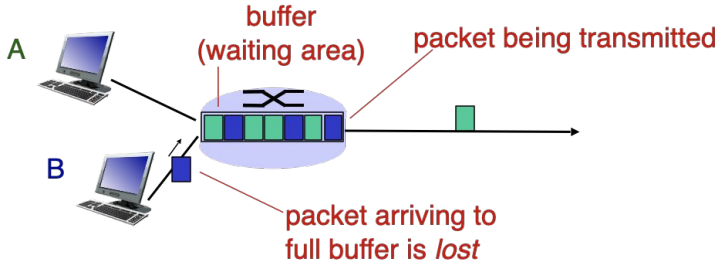
2. Now, suppose one way delay between the sender and receiver is 20 ms. Suppose sender needs to wait for an ACK after sending every 1KB packet. Assume ACK takes 20 ms to come back.What is the time required to send 1KB packet? What is the throughput?

Packet switching: How do packet delay or packet losses occur?



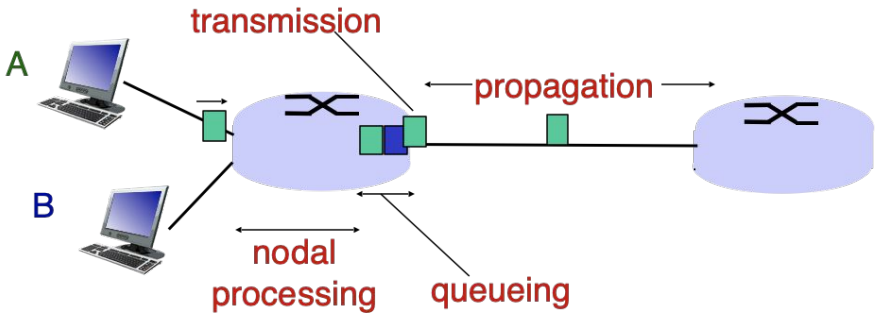
Packet arriving rate > Output link's capacity (temporarily)

- Incoming packet stored in queue (router's buffers) waiting for its turn



- Finite buffer capacity at the router
- Packet arrives when queue is full; packet is dropped
- Lost packet may or may not be retransmitted
  - Retransmission by previous node
  - Retransmission by source

Sources of packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

- **Transmission delay (d_{trans})**
  - Time required to place the entire packet over the output link
  - Depends on link capacity (bandwidth)
- **Propagation delay (d_{prop})**
  - Time required for one bit to propagate from one end to another
  - Depends on transmission medium characteristics
- **Nodal processing delay (d_{proc})**
  - Check bit errors
  - Determine output link
  - Depends on node specs
- **Queueing delay (d_{queue})**
  - Waiting time at output link for congestion
  - Depends on congestion level

Sources of packet delay (contd.)

- Transmission delay
  - Link data rate = R bits/sec
  - Packet size = L bits
  - Transmission delay, $\mathbf{d_{trans}}$ = L/R seconds
  - Example: Packet size, L = 1000 bits, Link data rate = 10^9 bps; $\mathbf{d_{trans}}$ = 1 us
- Propagation delay
  - Length of the link / distance travelled = d bits
  - Speed at which signal travels = s meters/sec
  - Propagation delay, $\mathbf{d_{prop}}$ = d/s seconds
  - Example: Speed of light, s = 3x10^8 m/s; d = 300m; $\mathbf{d_{prop}}$ = 1 us

Relation between the link utilization & propagation/transmission delays

$$d_{prop} < d_{trans}$$

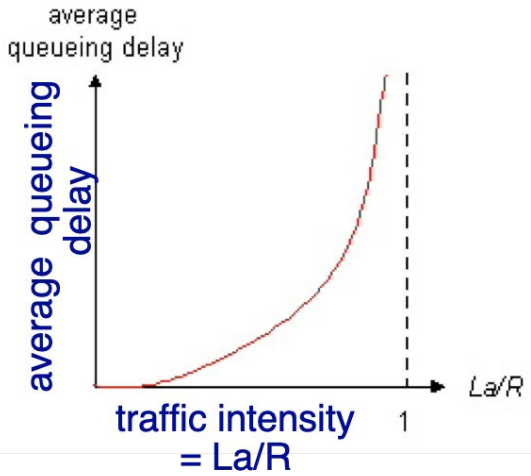
- The first bit reaches the receiver before the sender finishes transmission
- The data pipe is FULL

$$d_{prop} > d_{trans}$$

- Sender finishes transmission; receiver has not received even the first bit
- Typical scenario in case of long distance links

- **Limiting factor: How fast the link is?**
- **Design packet size for better network utilization (pipe should be FULL)**
- **Design protocols for better network utilization (pipe should be FULL)**

Queueing delay





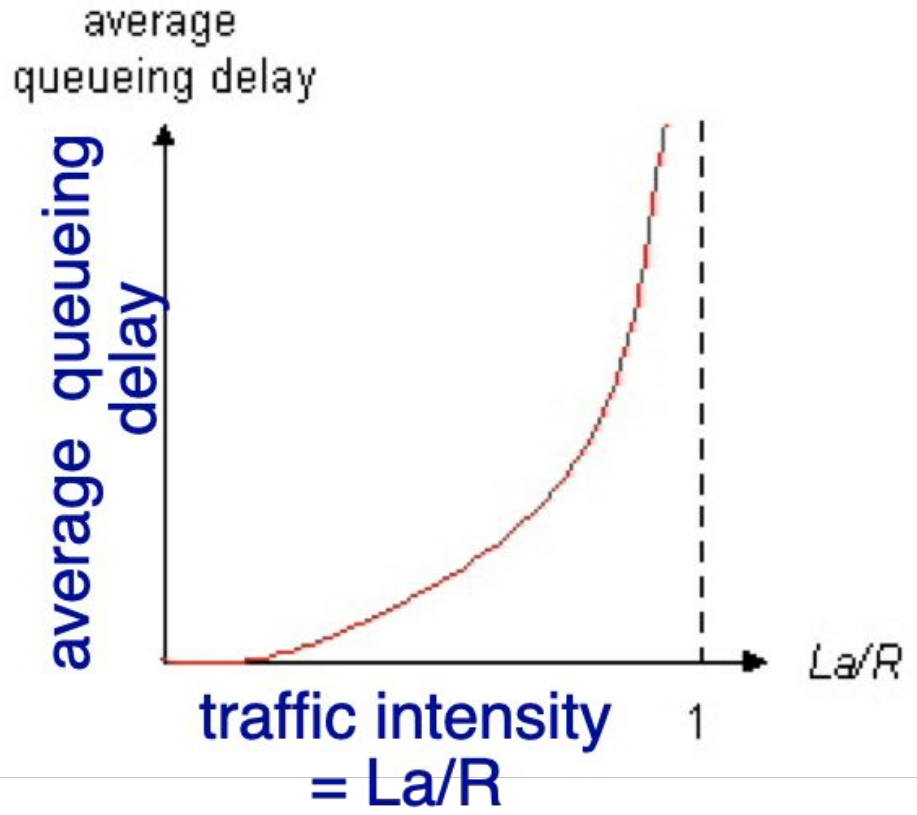
La/R ~ 0



La/R -> 1

R: Link bandwidth (bits per sec)
L: Packet length (bits)
a: average packet arrival rate (num_pkts per sec)

- **La/R ~ 0**
  - Small average queueing delay
- **La/R -> 1**
  - Large average queueing delay
- **La/R > 1**
  - Infinite average delay
    - More work; less capacity


average
queueing delay
average queueing delay
La/R
traffic intensity
= La/R
1

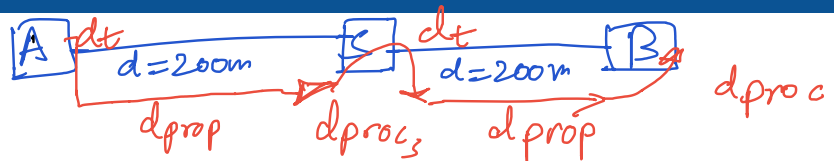
How do “real” Internet delays and losses look like?

Traceroute tool

You will study “traceroute” and many more network tools in your tutorial & programming assignment.

Concept of end to end delays

Problem 3 $s = 2 \times 10^8$ m/s, $L = 125B = 125*8b$, $DR = 1$Mbps



Consider a link A--S--B. A and B are end hosts and S is a switch/router in between. Each link has length 200m (speed of light in medium is 2x10^8 m/s). Each link has bandwidth of 1 Mbps. “A” has sent 125 byte packet. Each node (switch/end-host) takes 9 us to process each received packet. What is the **end-to-end delay** for this 125 byte packet **in microseconds**?

$t = d_{trans_A} + d_{prop} + d_{proc_S} + d_{trans_S} + d_{prop} + d_{proc_B}$

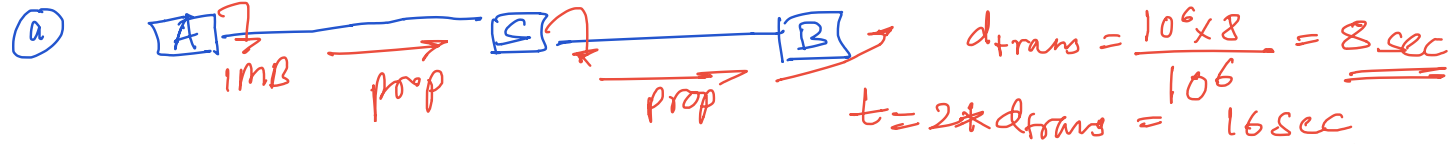
$= 2020\,\mu s$

$d_{trans} = \frac{L}{DR} = \frac{125*8}{10^6} = 1000\,\mu s$

$d_{prop} = \frac{l}{s} = \frac{200}{2 \times 10^8} = 1\,\mu s$

Benefits of packetization in store-and-forward networks

Problem 4



Consider a link A--S--B problem **(problem 3).** Suppose "A" wants to send a 1MB file to "B". "A" has two options:

a. Send the entire file as one big packet on both the links
b. Break up the packet into 100 byte chunks, add a 25 byte header per chunk, and send it on the links.

#pkts $= \frac{10^6 B}{100B} = 10000$ pkts $L = 100*8+25 = 125B$

Ignore the propagation and processing delays, as they are very small.

$d_{trans} = \frac{125 \times 8}{10^6} = 1$ ms

Compute the **end-to-end delays** for both the cases.



$d_{10000\ pkts\ (trans)} = 10$ s

$= 10 + 0.001 = 10.001$ sec

Circuit switching vs. Packet switching delays

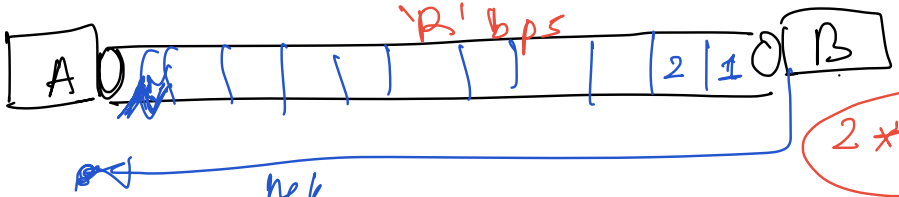
Problem 5

Consider a network with two end hosts and one switch in between, A--S--B. Each link has bandwidth of 1 Mbps. Suppose A and B wish to transfer a 1MB file between them. The switch can do circuit switching(CS) or packet switching(PS). **Ignore propagation delays.**

For CS: Setup request/response message = 1KB each; node processing delay = 2ms. After the circuit is setup, data is pumped at link rate.

For PS: Split a file into 100 byte segments and add 25 byte headers; ***ignore node processing delays (~ 0)*** → ***otherwise, t_proc > t_trans*** Use datagram approach of packet switching (i.e., No call setup involved)

1. **What are the delays and overheads (in "num bytes") for both, CS and PS?**
2. **What are the pros and cons of each?**



$Tpt = \frac{L}{time}$

2 * dprop

Round Trip Time (RTT)

waiting time = 0

Bandwidth Delay Product (BDP)

= Bottleneck Bandwidth * RTT

Q. How much data should the sender send to utilize the link efficiently, i.e., 100%

OR FILL THE NETWORK PIPE

Bandwidth Delay Product (BDP) calculation

Problem 6

A sender and receiver are communicating over a 1Mbps link. Sender sends **1KB packets**. Receiver sends **250 byte ACKs**. Suppose **round trip propagation delay** is **10 ms** (assume long distance link).

A. What is the throughput if sender sends 1 packet and waits for an ACK?

B. What is the bandwidth delay product (BDP) of the link?

C. How is BDP useful for network design?